

Relevance-Based Prediction: Factor Selection for Saugus Schools

Utilizing Czasonis, Kritzman & Turkington (2024) — Applied to MA School District Analysis

Four outcomes: MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, Education Budget Share

RBP run once with ALL candidates — Exhibit 5 importance scores each feature's effect

Per Kritzman fn. 12: positive = helps, ≈ 0 = benign (diversified away), < 0 = harmful

Saugus analyzed as prediction task; most/least relevant towns identified per Exhibit 4

Leave-one-out LOO r validates the retained feature set across all MA districts

Retained Feature Counts			
MCAS Grades 3-8	Dropout Rate	MCAS Grade 10 (ELA)	Education Budget Share
Candidates: 17 features	Candidates: 18 features	Candidates: 18 features	Candidates: 18 features
Retained (imp > 0): 17 of 17	Retained (imp > 0): 17 of 18	Retained (imp > 0): 17 of 18	Retained (imp > 0): 18 of 18
LOO r = +0.945	LOO r = +0.839	LOO r = +0.899	LOO r = +0.901

Pre-Specified Candidate Pool — Feature Reduction

Features selected on domain grounds before running RBP. Reduces K/N ratio from ~0.19 to ~0.09 (target ≈ 0.085).

Kept — 14 pure predictors

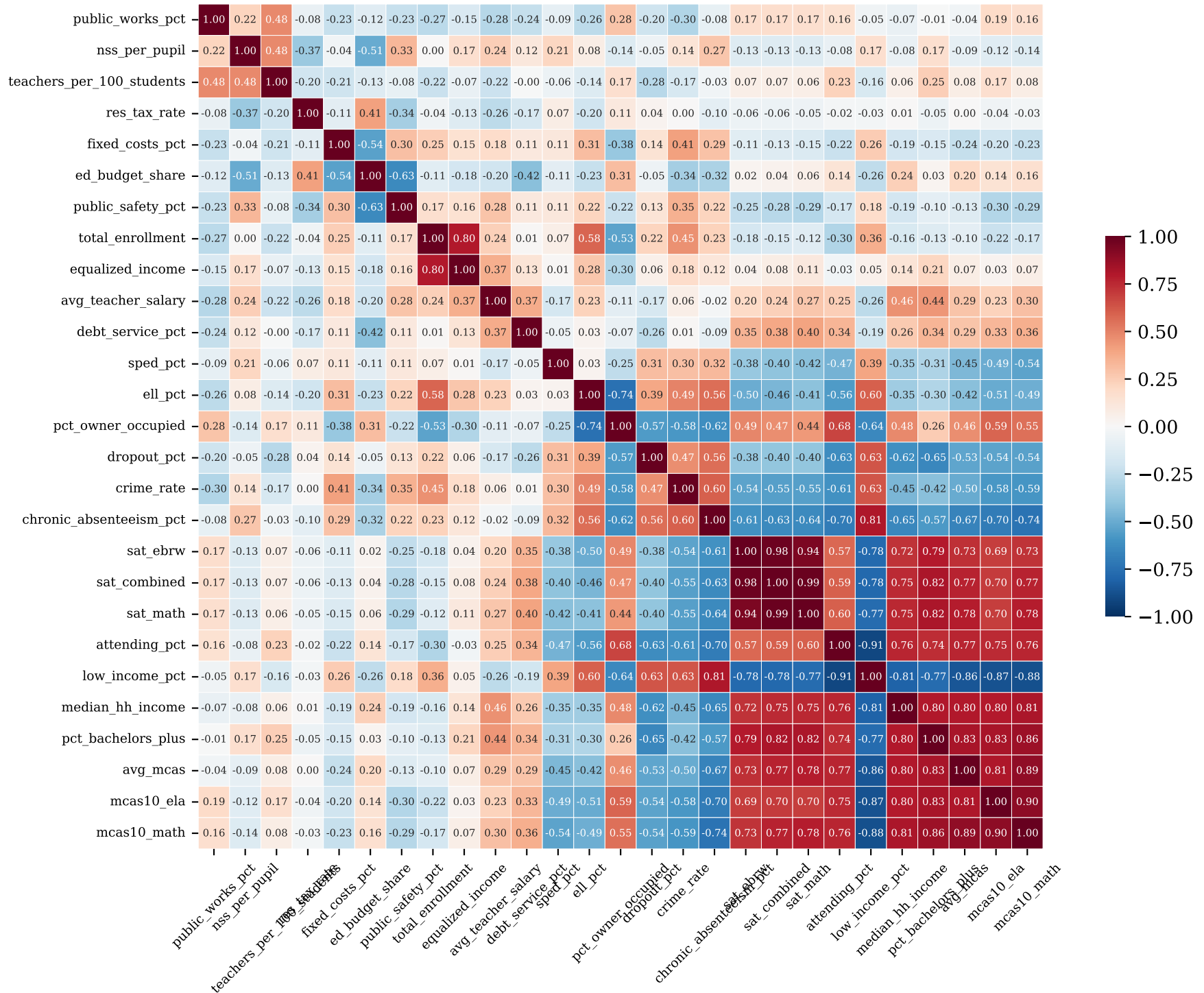
Group	Feature	Measures
Poverty	low_income_pct	% low-income students
Wealth	median_hh_income	Median household income
Wealth	equalized_income	Equalized property value/capita
Human capital	pct_bachelors_plus	% adults with bachelor's+
Housing	pct_owner_occupied	% owner-occupied homes
Community	crime_rate	Crime incidents per capita
Community	res_tax_rate	Residential tax rate
Engagement	chron_absenteeism_pct	% chronically absent
Demographics	ell_pct	% English language learners
Demographics	sped_pct	% special education students
Size	total_enrollment	District enrollment
Staffing	teachers/100_students	Teachers per 100 students
Staffing	avg_teacher_salary	Average teacher salary
Spending	nss_per_pupil	Net school spending/pupil

Dropped — 12 redundant/problematic features

Dropped	Kept instead	Reason
high_needs_pct	low_income_pct	$r=+0.981$ — near-duplicate
acs_poverty_pct	low_income_pct	$r=+0.87$ — redundant poverty
foundation_budget_pp	nss_per_pupil	$r=+0.92$ — redundant spending
ch70_per_pupil	equalized_income	$r\approx-0.87$ — inverse wealth proxy
teacher_fte	total_enrollment	$r=+0.994$ — linear with enroll
total_population	total_enrollment	$r=+0.953$ — size proxy
teachers_per_100_fte	t_per_100_students	$r=+0.872$ — duplicate ratio
teacher_spending_pp	nss_per_pupil	$r\approx+0.85$ — redundant spending
gf_exp_per_capita	res_tax_rate	$r\approx+0.75$ — redundant municipal
pct_65_plus	—	Weak signal, no policy lever
com_tax_rate	res_tax_rate	$r\approx+0.65$ — commercial tax dup
mcas10_math	mcas10_elar	$r=+0.90$ — one grade-10 measure

Correlation threshold for deduplication: $|r| > 0.87$. Cross-outcome features (avg_mcas, mcas10_ela, dropout_pct, attending_pct) handled separately per model via exclusion rules.

Feature Correlation Matrix — All Candidates (hierarchically clustered)
Features ordered by |correlation| — dark red/blue clusters = high redundancy

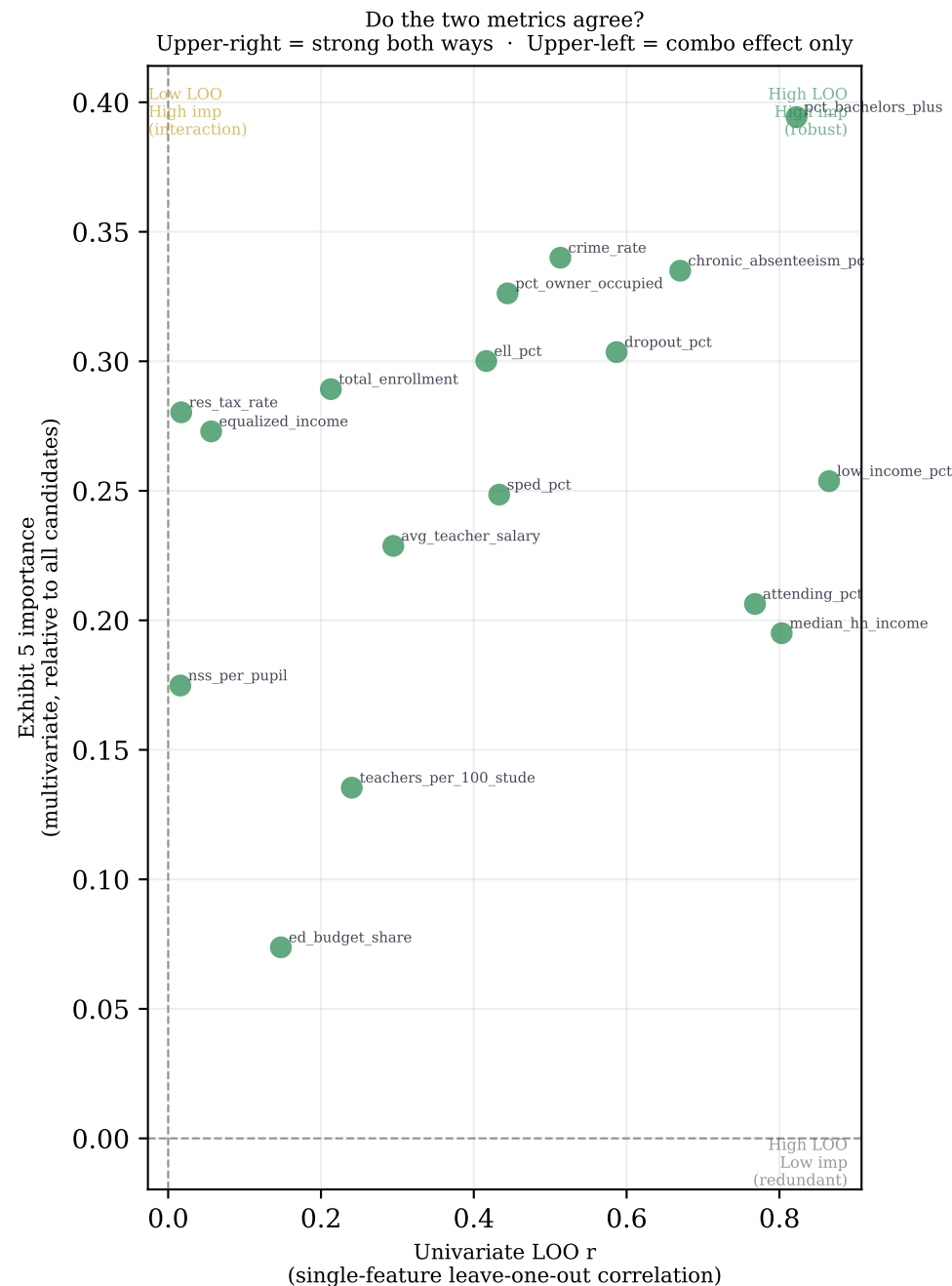
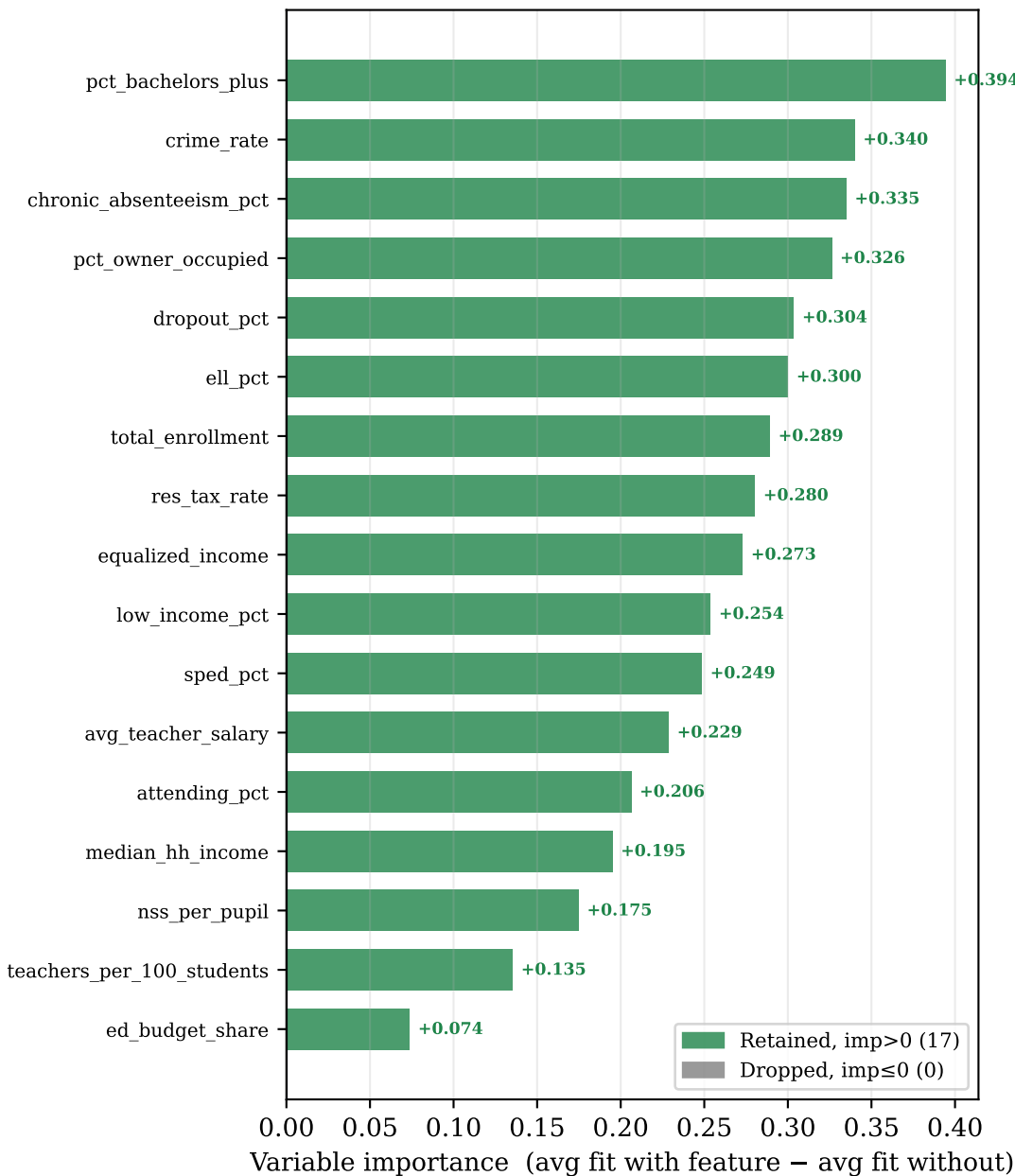


Pearson r. Hierarchical average-linkage clustering on $|1-r|$ distance. Features with $|r| > 0.87$ collapsed to one representative in the candidate pool.

Variable Importance (Exhibit 5): MCAS Grades 3-8

17 candidates · n_random=1000 · 17 retained (imp > 0) · 0 with imp ≤ 0 (benign or harmful)

All Candidates — Variable Importance
Green = helps (>0) · Grey ≈ 0 benign · Red < 0 harmful

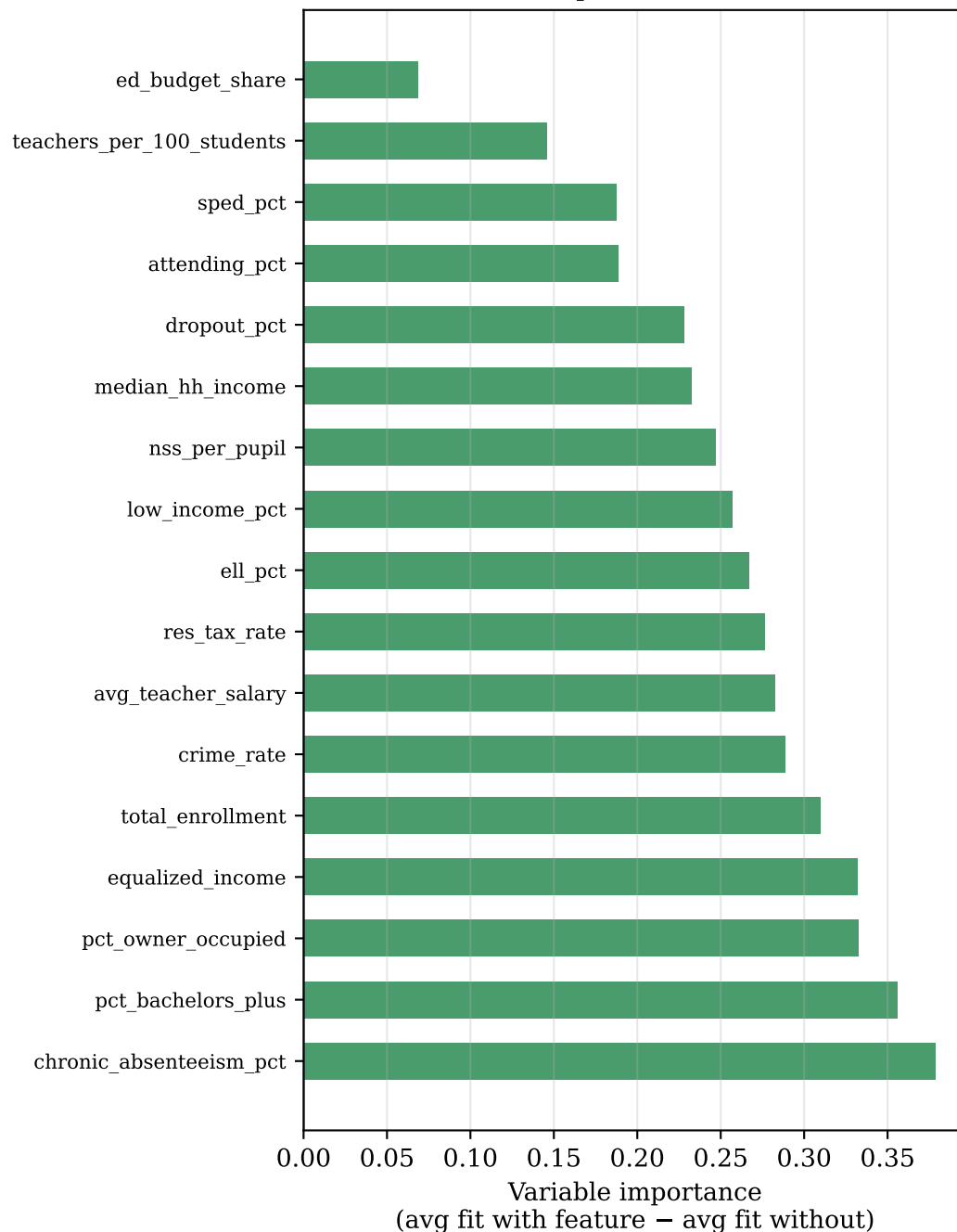


Variable importance: avg adjusted fit of grid cells containing feature – avg fit of cells without it. Univariate LOO r: single-feature leave-one-out correlation.

Saugus RBP Analysis: MCAS Grades 3-8

Prediction task = Saugus · Predicted: 22.3pp Actual: 29.0pp Gap: +6.7pp (Fit: 0.1469)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	avg_mcas
Everett	0.0773	0.18
Gardner	0.0677	0.21
Pittsfield	0.0660	0.24
Dracut	0.0609	0.38
Agawam	0.0549	0.34

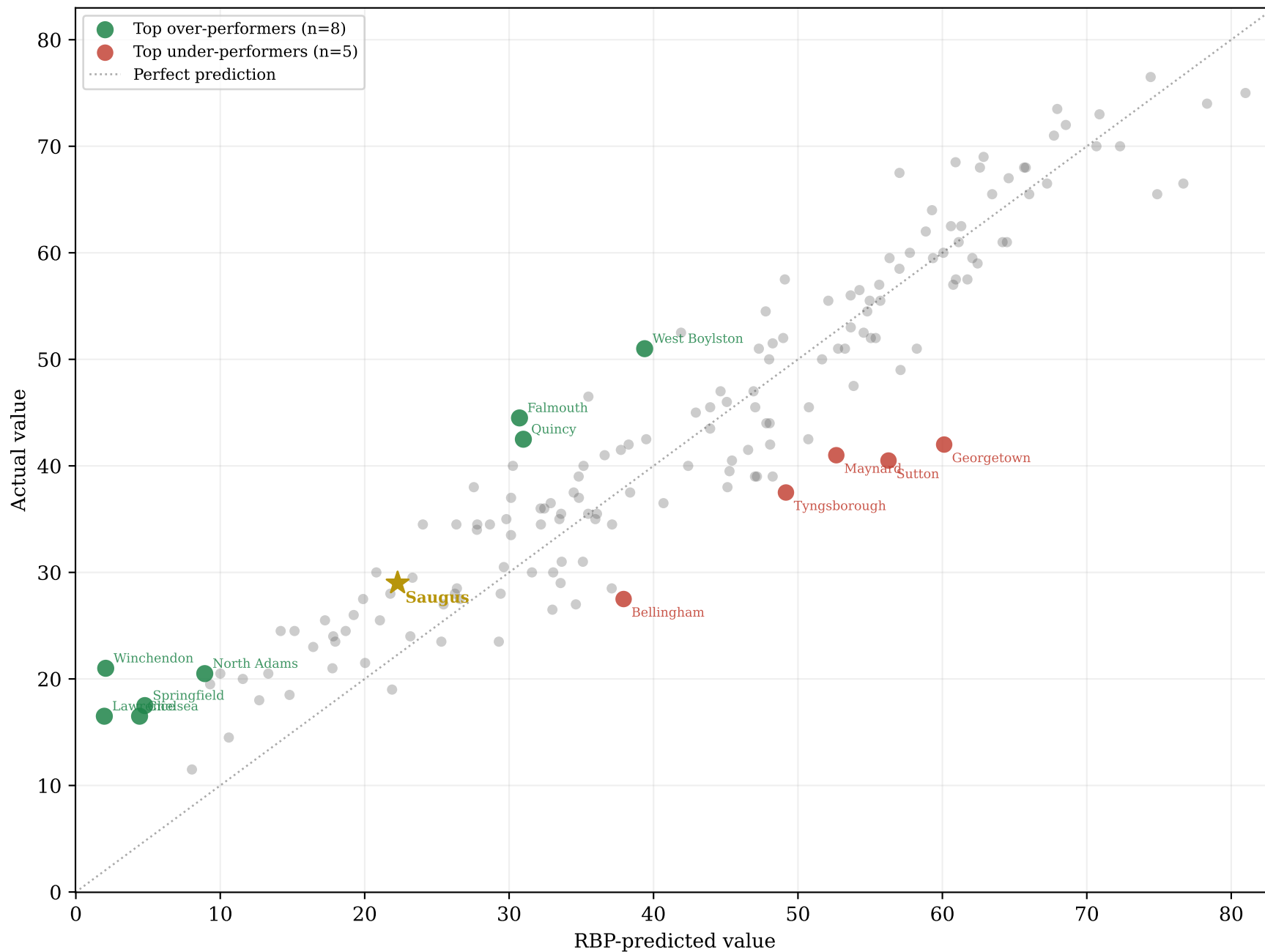
Least Relevant Towns (lowest weight)

Town	Weight	avg_mcas
Andover	-0.0182	0.66
Hingham	-0.0178	0.73
Foxborough	-0.0177	0.51
Marshfield	-0.0170	0.55
Ipswich	-0.0169	0.56

Over- and Under-Performers: MCAS Grades 3-8

Saugus gap: +6.7pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — MCAS Grades 3-8 (LOO r = 0.945)



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: MCAS Grades 3-8

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Winchendon	Lawrence	Falmouth	Imp
chronic_absenteeism_pct	31.2	35.6	29.2	24.0	+0.379
pct_bachelors_plus	31.4	23.9	16.4	51.9	+0.356
pct_owner_occupied	73.9	77.8	30.3	82.7	+0.333
equalized_income	1.17B	278M	1.77B	1.68B	+0.332
total_enrollment	3,084	1,360	15,500	3,130	+0.310
crime_rate	2,813	3,287	3,435	3,500	+0.289
avg_teacher_salary	89,778	83,722	84,516	92,617	+0.282
res_tax_rate	10.7	12.5	9.25	6.28	+0.276
ell_pct	13.0	1.90	42.1	7.50	+0.267
low_income_pct	46.6	56.8	85.1	39.5	+0.257
nss_per_pupil	17,369	19,559	23,178	25,271	+0.247
median_hh_income	100,819	85,160	57,903	92,862	+0.233
dropout_pct	2.90	7.50	7.80	1.00	+0.228
attending_pct	64.4	36.4	43.4	68.0	+0.189
sped_pct	18.8	27.9	19.4	24.4	+0.188

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	29.0	22.3	+6.7pp
Winchendon	21.0	2.1	+18.9pp
Lawrence	16.5	2.0	+14.5pp
Falmouth	44.5	30.7	+13.8pp
Springfield	17.5	4.8	+12.7pp
Chelsea	16.5	4.4	+12.1pp
West Boylston	51.0	39.4	+11.6pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grades 3-8

Predicted 22.3pp · Actual 29.0pp · Gap +6.7pp (higher is better) · Saugus ranks 30 of 168 MA districts on this measure

The Bottom Line

Saugus scores 29.0% on MCAS grades 3–8 (ELA + Math, meeting/exceeding), almost exactly its demographic prediction of 28.7% — a +0.3pp gap that is essentially on target. Saugus ranks 103rd of 169 MA districts, around the middle of the pack. This is the “baseline” result: a district with Saugus's income, poverty, and enrollment profile is expected to land here, and it does.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.38
Adults with a bachelor's degree+	31.4%	46.9%	+0.36
Owner-occupied housing units	73.9%	78.3%	+0.33
Equalized property value per capita	\$1.17B	\$935M	+0.33
Total district enrollment	3,084	1,669	+0.31

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	29.8%	-1.4pp
Adults with a bachelor's degree+	31.4%	25.5%	-6.0pp
Owner-occupied housing units	73.9%	51.3%	-22.6pp
Equalized property value per capita	\$1.17B	\$1.29B	+119M
Total district enrollment	3,084	5,212	+2,128

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

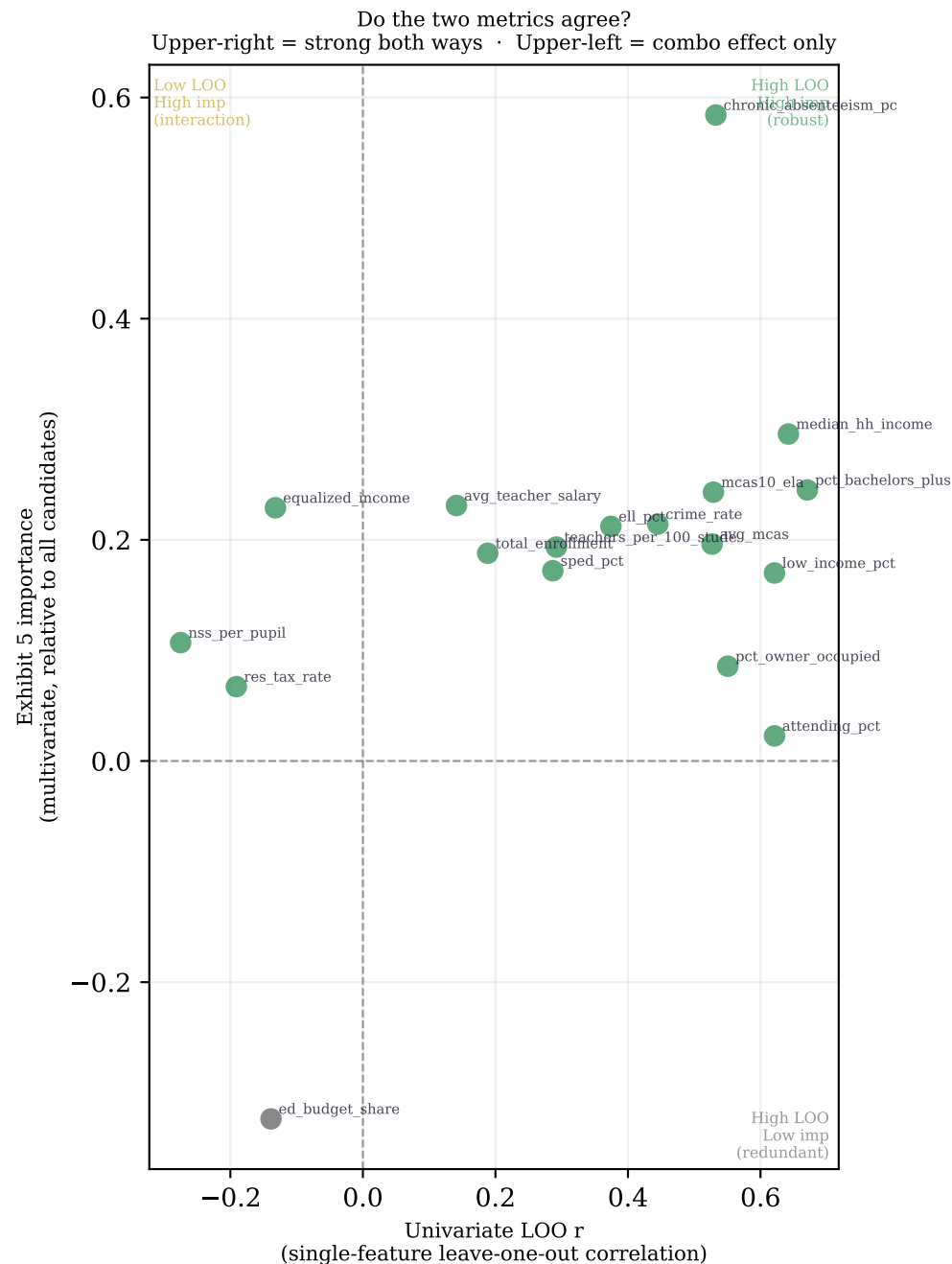
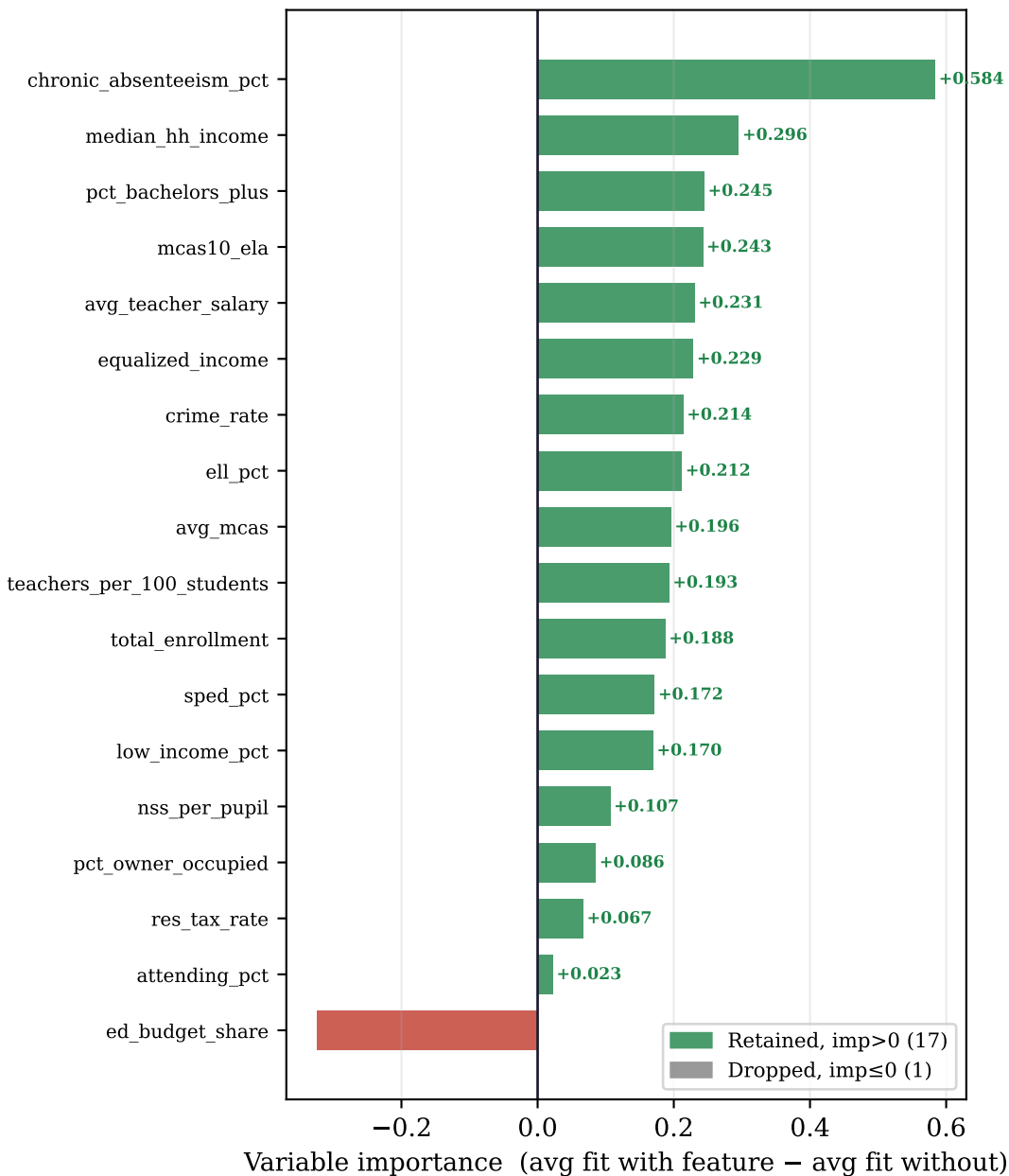
The Takeaway

Because Saugus is on target here, the importance scores point less to “what's wrong” and more to “what would move the needle.” Chronic absenteeism (31.2%, nearly double the 15.8% state median) is the single largest local driver — and the largest gap versus over-performing districts. Closing even part of that gap is the most plausible route from an average result to an above-average one.

Variable Importance (Exhibit 5): Dropout Rate

18 candidates · n_random=1000 · 17 retained (imp > 0) · 1 with imp ≤ 0 (benign or harmful)

All Candidates — Variable Importance
Green = helps (>0) · Grey ≈ 0 benign · Red < 0 harmful

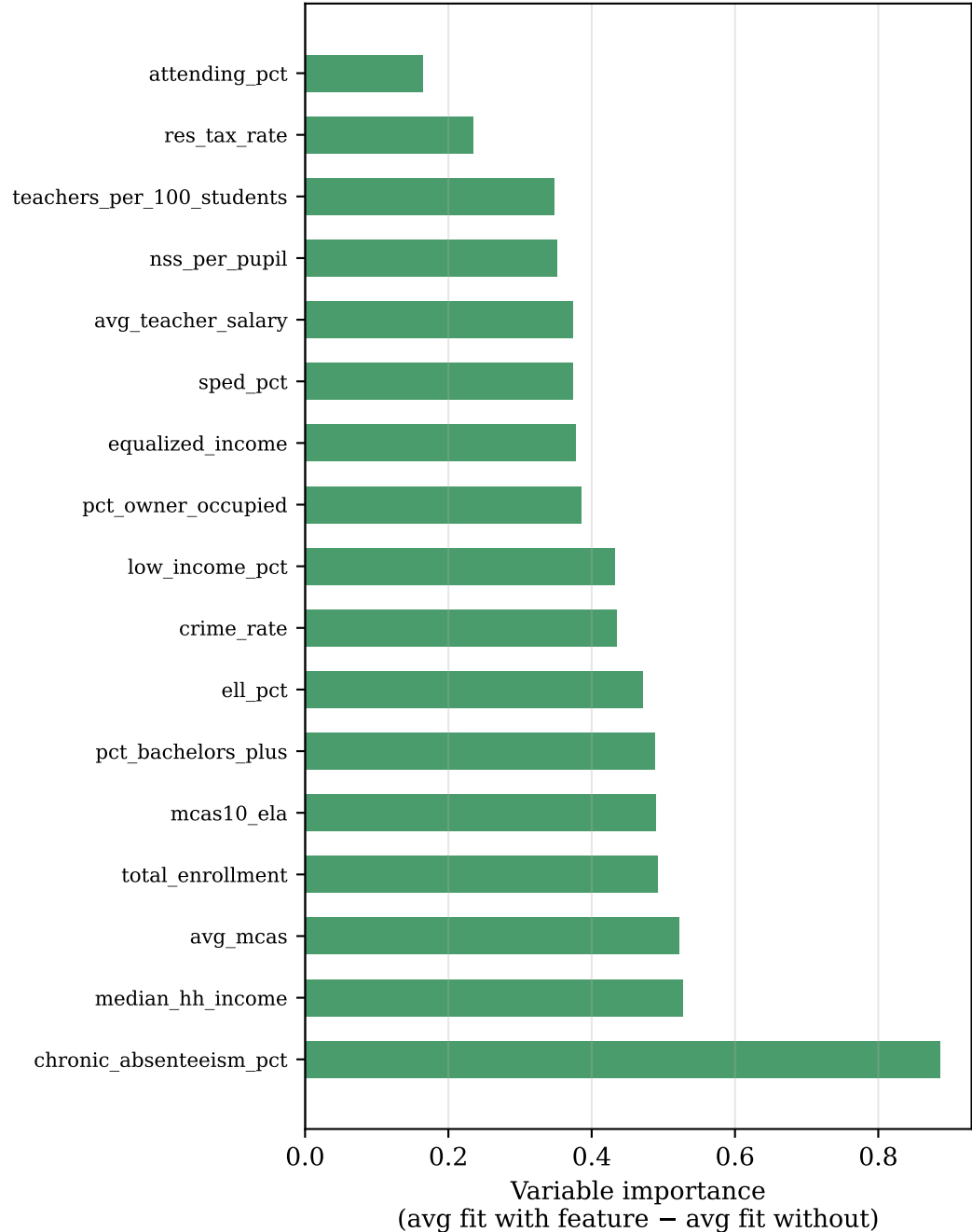


Variable importance: avg adjusted fit of grid cells containing feature – avg fit of cells without it. Univariate LOO r: single-feature leave-one-out correlation.

Saugus RBP Analysis: Dropout Rate

Prediction task = Saugus · Predicted: 3.6pp Actual: 2.9pp Gap: -0.7pp (Fit: 0.1840)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	dropout_pct
Winchendon	0.0655	7.50
Dracut	0.0566	1.40
Everett	0.0538	5.20
Agawam	0.0531	1.30
Randolph	0.0531	3.30

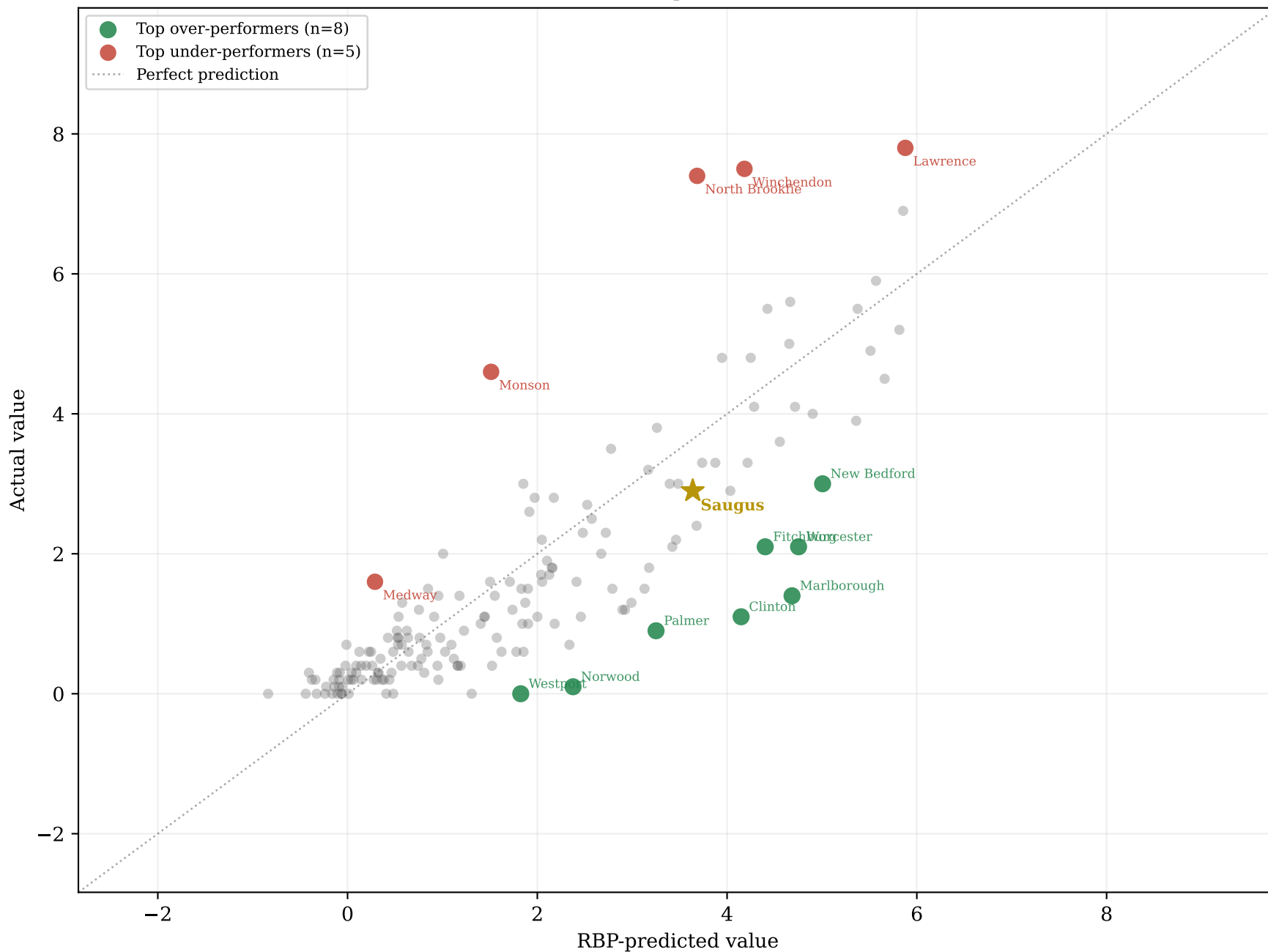
Least Relevant Towns (lowest weight)

Town	Weight	dropout_pct
Dedham	-0.0151	0.40
Andover	-0.0138	0.20
Hatfield	-0.0137	0.00
Natick	-0.0134	0.70
North Reading	-0.0131	0.00

Over- and Under-Performers: Dropout Rate

Saugus gap: -0.7pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — Dropout Rate (LOO r = 0.839)



Residual = actual - predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: Dropout Rate

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Marlborou	Clinton	Worcester	Imp
chronic_absenteeism_pct	31.2	23.5	7.70	28.3	+0.886
median_hh_income	100,819	95,047	93,849	67,544	+0.527
avg_mcas	0.29	0.24	0.30	0.23	+0.523
total_enrollment	3,084	5,572	2,237	27,606	+0.493
mcas10_ela	0.47	0.40	0.43	0.40	+0.490
pct_bachelors_plus	31.4	40.6	35.1	34.1	+0.488
ell_pct	13.0	32.9	19.7	30.4	+0.471
crime_rate	2,813	3,695	324.5	5,437	+0.435
low_income_pct	46.6	58.3	52.7	71.4	+0.432
pct_owner_occupied	73.9	56.2	52.9	42.4	+0.385
equalized_income	1.17B	1.65B	514M	5.00B	+0.378
sped_pct	18.8	16.4	21.9	21.9	+0.374
avg_teacher_salary	89,778	87,213	87,059	101,529	+0.374
nss_per_pupil	17,369	20,169	18,512	22,244	+0.352
teachers_per_100_student	5.87	7.36	7.31	7.40	+0.347

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	2.9	3.6	-0.7pp
Marlborough	1.4	4.7	+3.3pp better
Clinton	1.1	4.1	+3.0pp better
Worcester	2.1	4.8	+2.7pp better
Palmer	0.9	3.3	+2.4pp better
Fitchburg	2.1	4.4	+2.3pp better
Norwood	0.1	2.4	+2.3pp better

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Dropout Rate

Predicted 3.6pp · Actual 2.9pp · Gap -0.7pp (lower is better) · Saugus ranks 41 of 169 MA districts on this measure

The Bottom Line

Saugus's dropout rate is 2.9%, a full 0.8pp better than its predicted 3.7% (lower is better for this measure) — ranking 34th of 169 districts, in the top 20%. Of the four outcomes, this is Saugus's clearest over-performance relative to demographic expectation.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.89
Median household income	\$100,819	\$109,970	+0.53
MCAS 3–8 (% meeting/exceeding)	29.0%	40.0%	+0.52
Total district enrollment	3,084	1,669	+0.49
MCAS 10 ELA (% meeting/exceeding)	47.0%	61.0%	+0.49

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	24.3%	-6.9pp
Median household income	\$100,819	\$86,319	-14,500
MCAS 3–8 (% meeting/exceeding)	29.0%	26.5%	-2.5pp
Total district enrollment	3,084	4,668	+1,584
MCAS 10 ELA (% meeting/exceeding)	47.0%	40.5%	-6.5pp

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

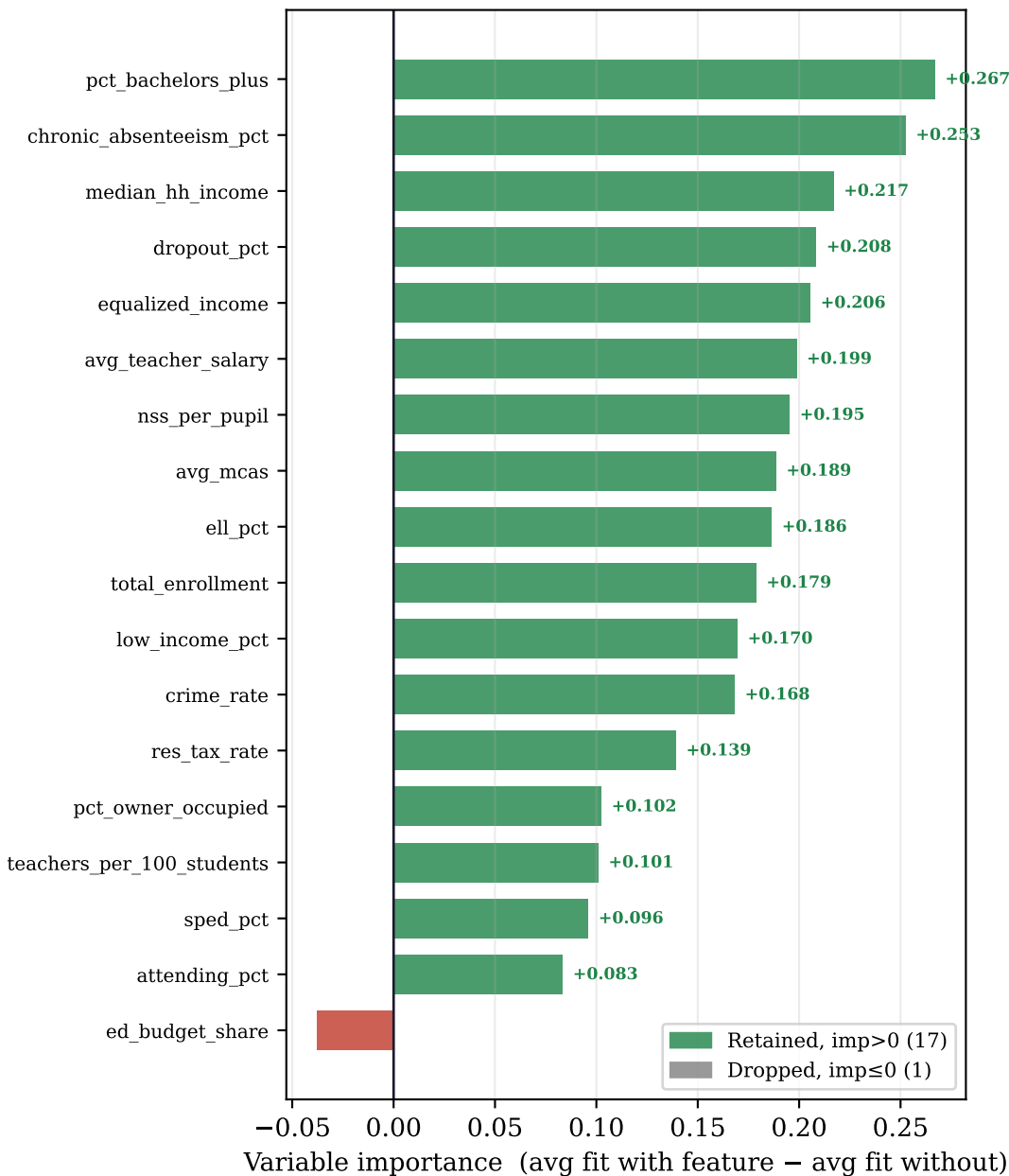
The Takeaway

Chronic absenteeism is overwhelmingly the strongest local driver here — more than double any other factor — yet Saugus still keeps fewer students from dropping out than a district with this absenteeism rate typically would. That gap between “high absenteeism” and “low dropout” suggests Saugus's re-engagement and attendance-recovery efforts are working — and that rising absenteeism is a real risk to this result if it isn't addressed.

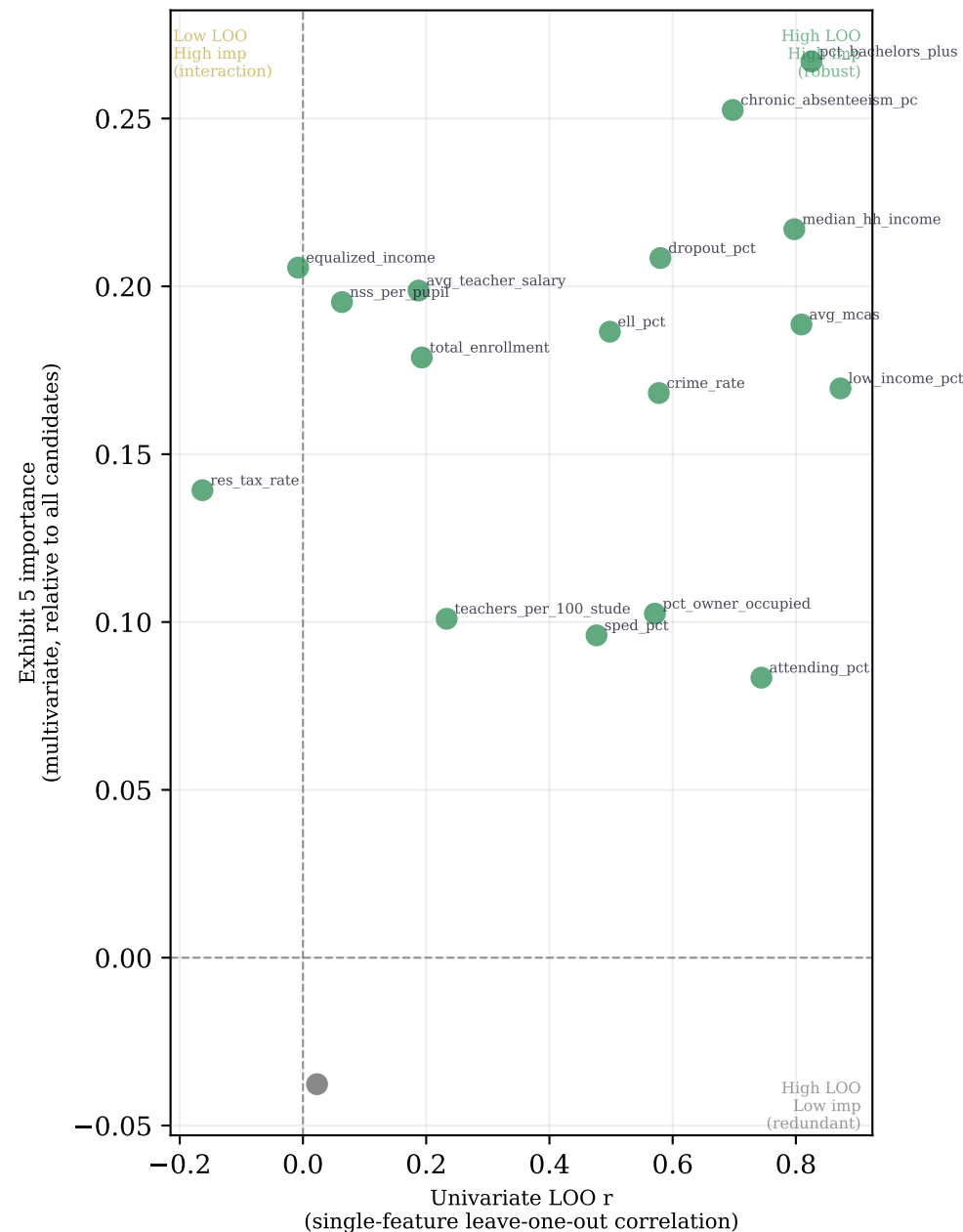
Variable Importance (Exhibit 5): MCAS Grade 10 (ELA)

18 candidates · n_random=1000 · 17 retained (imp > 0) · 1 with imp ≤ 0 (benign or harmful)

All Candidates — Variable Importance
Green = helps (>0) · Grey ≈ 0 benign · Red < 0 harmful



Do the two metrics agree?
Upper-right = strong both ways · Upper-left = combo effect only

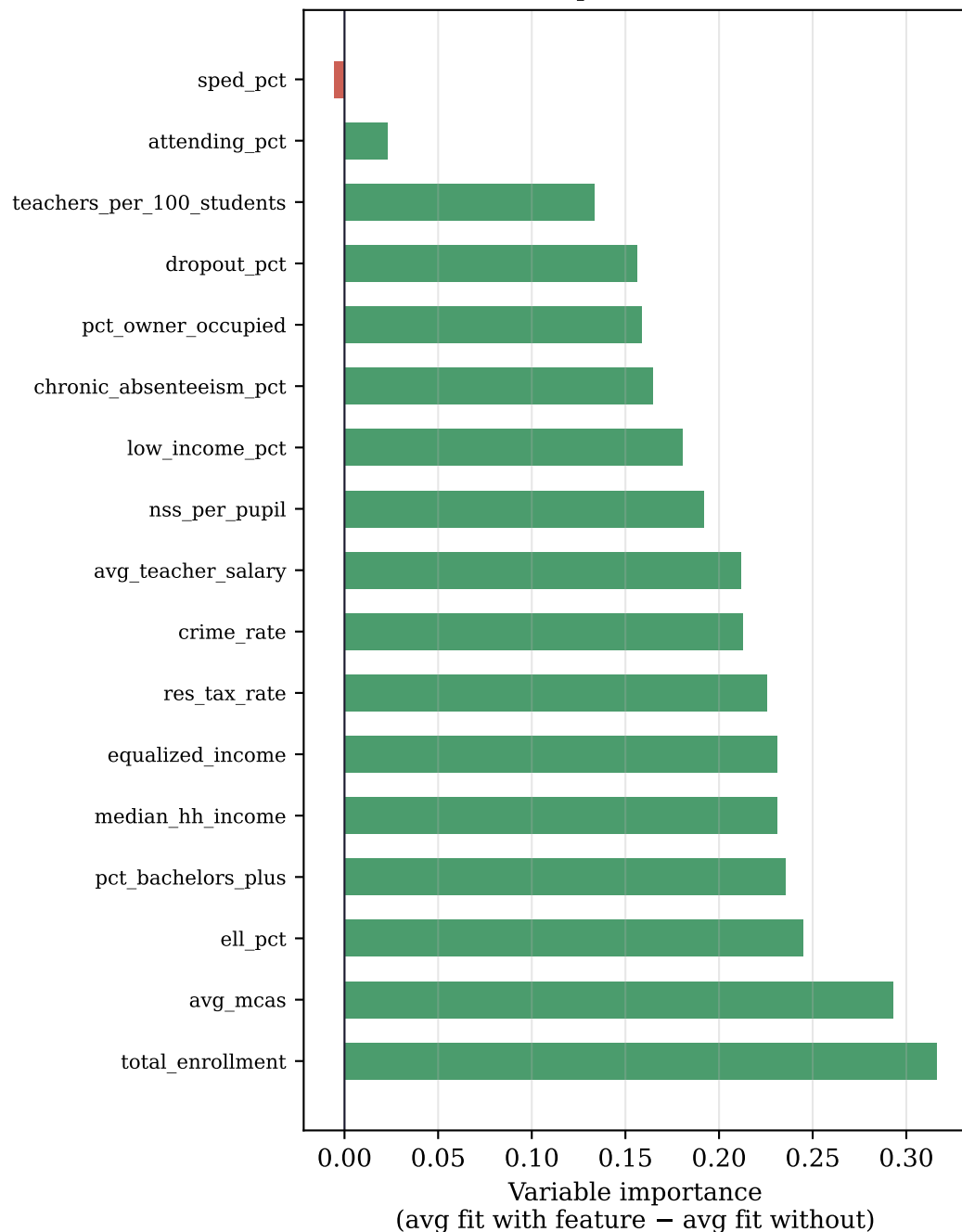


Variable importance: avg adjusted fit of grid cells containing feature – avg fit of cells without it. Univariate LOO r: single-feature leave-one-out correlation.

Saugus RBP Analysis: MCAS Grade 10 (ELA)

Prediction task = Saugus · Predicted: 42.3pp Actual: 47.0pp Gap: +4.7pp (Fit: 0.1573)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

Town	Weight	mcas10_ela
Dracut	0.0544	0.53
Randolph	0.0482	0.36
Everett	0.0452	0.37
Gardner	0.0446	0.40
Agawam	0.0446	0.54

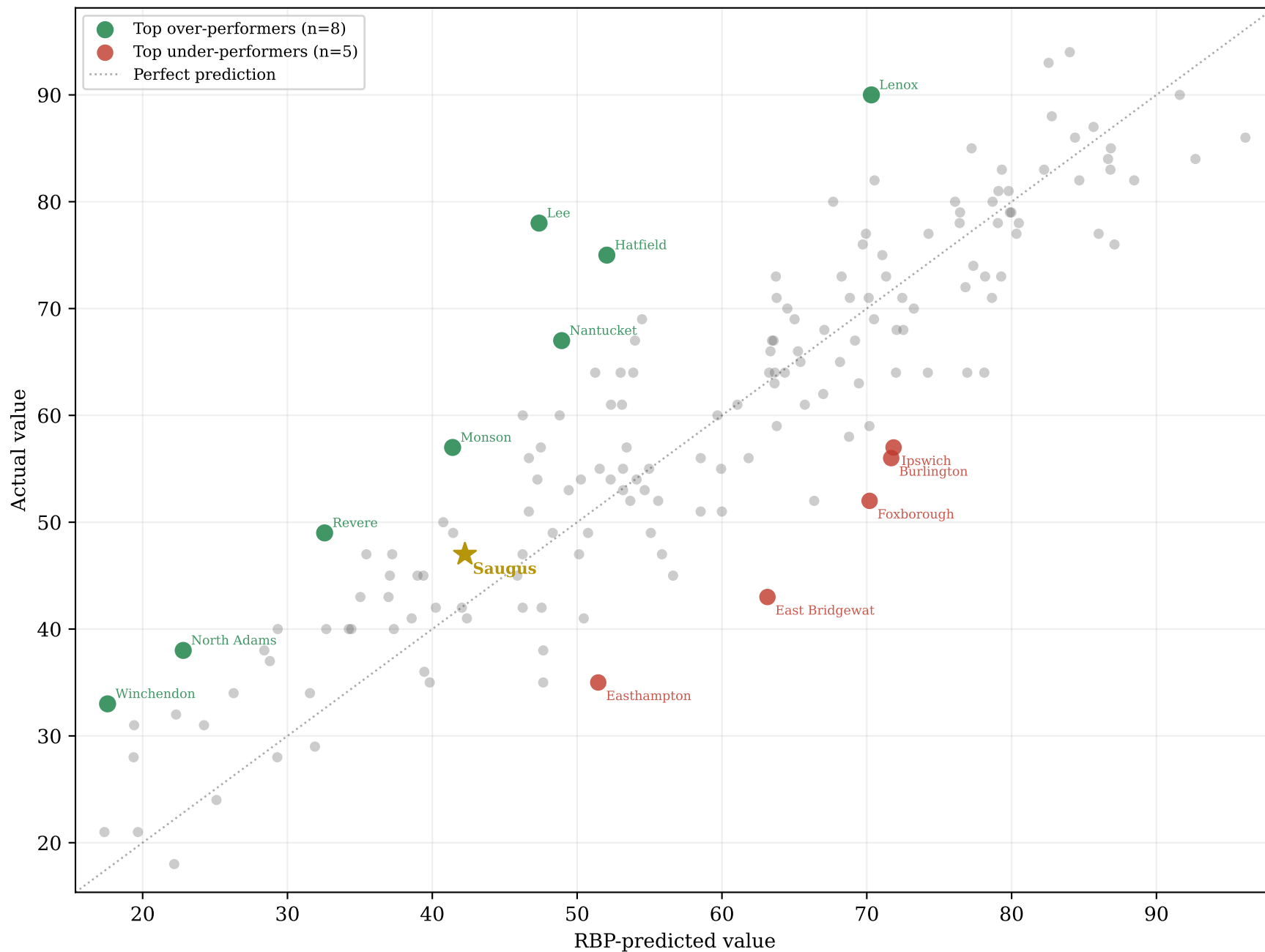
Least Relevant Towns (lowest weight)

Town	Weight	mcas10_ela
Dedham	-0.0132	0.59
Cohasset	-0.0129	0.93
Andover	-0.0127	0.79
Bedford	-0.0122	0.83
Natick	-0.0120	0.72

Over- and Under-Performers: MCAS Grade 10 (ELA)

Saugus gap: +4.7pp · Gold star = Saugus · Green = over-performers · Red = under-performers

Actual vs Predicted — MCAS Grade 10 (ELA) (LOO r = 0.899)



Residual = actual – predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: MCAS Grade 10 (ELA)

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Lee	Hatfield	Lenox	Imp
total_enrollment	3,084	746.0	412.0	814.0	+0.316
avg_mcas	0.29	0.34	0.37	0.57	+0.293
ell_pct	13.0	5.60	0.30	4.10	+0.245
pct_bachelors_plus	31.4	45.1	48.7	49.5	+0.235
median_hh_income	100,819	74,222	100,381	97,569	+0.231
equalized_income	1.17B	210M	132M	310M	+0.231
res_tax_rate	10.7	11.2	15.3	9.07	+0.226
crime_rate	2,813	3,348	1,442	2,695	+0.213
avg_teacher_salary	89,778	82,282	87,392	84,669	+0.212
nss_per_pupil	17,369	24,155	24,753	23,945	+0.192
low_income_pct	46.6	41.2	20.5	24.3	+0.181
chronic_absenteeism_pct	31.2	13.3	13.3	18.3	+0.165
pct_owner_occupied	73.9	78.4	74.8	71.3	+0.159
dropout_pct	2.90	1.00	0.00	0.80	+0.156
teachers_per_100_student	5.87	9.69	9.59	10.1	+0.133

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	47.0	42.3	+4.7pp
Lee	78.0	47.4	+30.6pp
Hatfield	75.0	52.1	+22.9pp
Lenox	90.0	70.3	+19.7pp
Nantucket	67.0	48.9	+18.1pp
Revere	49.0	32.6	+16.4pp
Monson	57.0	41.4	+15.6pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grade 10 (ELA)

Predicted 42.3pp · Actual 47.0pp · Gap +4.7pp (higher is better) · Saugus ranks 52 of 169 MA districts on this measure

The Bottom Line

Saugus's grade 10 ELA proficiency is 47.0%, 4.7pp above its predicted 42.3% — ranking 54th of 169 districts, in the top third. This is Saugus's strongest relative academic result of the four outcomes.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Total district enrollment	3,084	1,669	+0.32
MCAS 3-8 (% meeting/exceeding)	29.0%	40.0%	+0.29
English language learners	13.0%	3.4%	+0.24
Adults with a bachelor's degree+	31.4%	46.9%	+0.24
Median household income	\$100,819	\$109,970	+0.23

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Total district enrollment	3,084	1,135	-1,949
MCAS 3-8 (% meeting/exceeding)	29.0%	26.5%	-2.5pp
English language learners	13.0%	3.0%	-10.0pp
Adults with a bachelor's degree+	31.4%	37.1%	+5.6pp
Median household income	\$100,819	\$84,614	-16,206

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

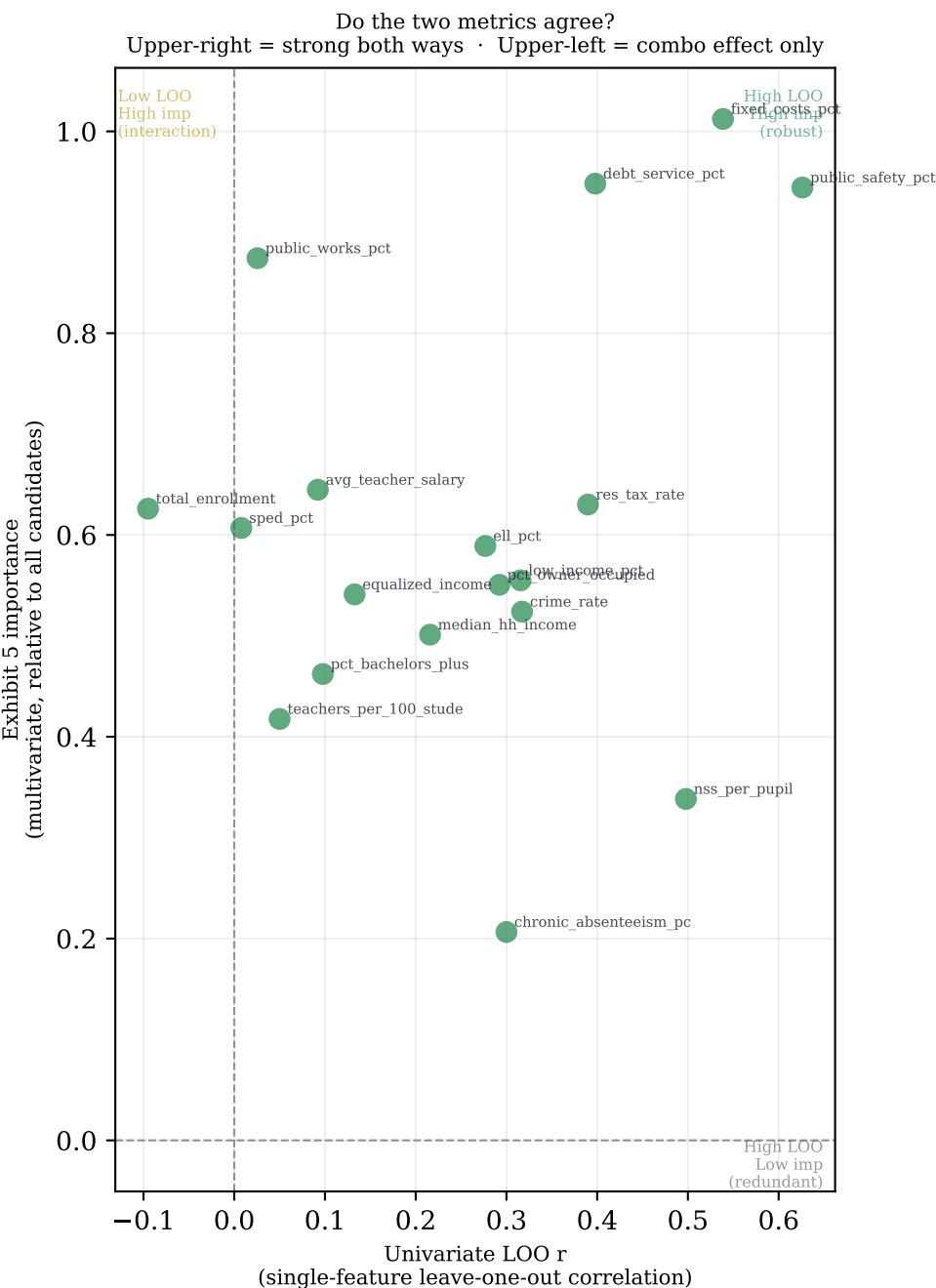
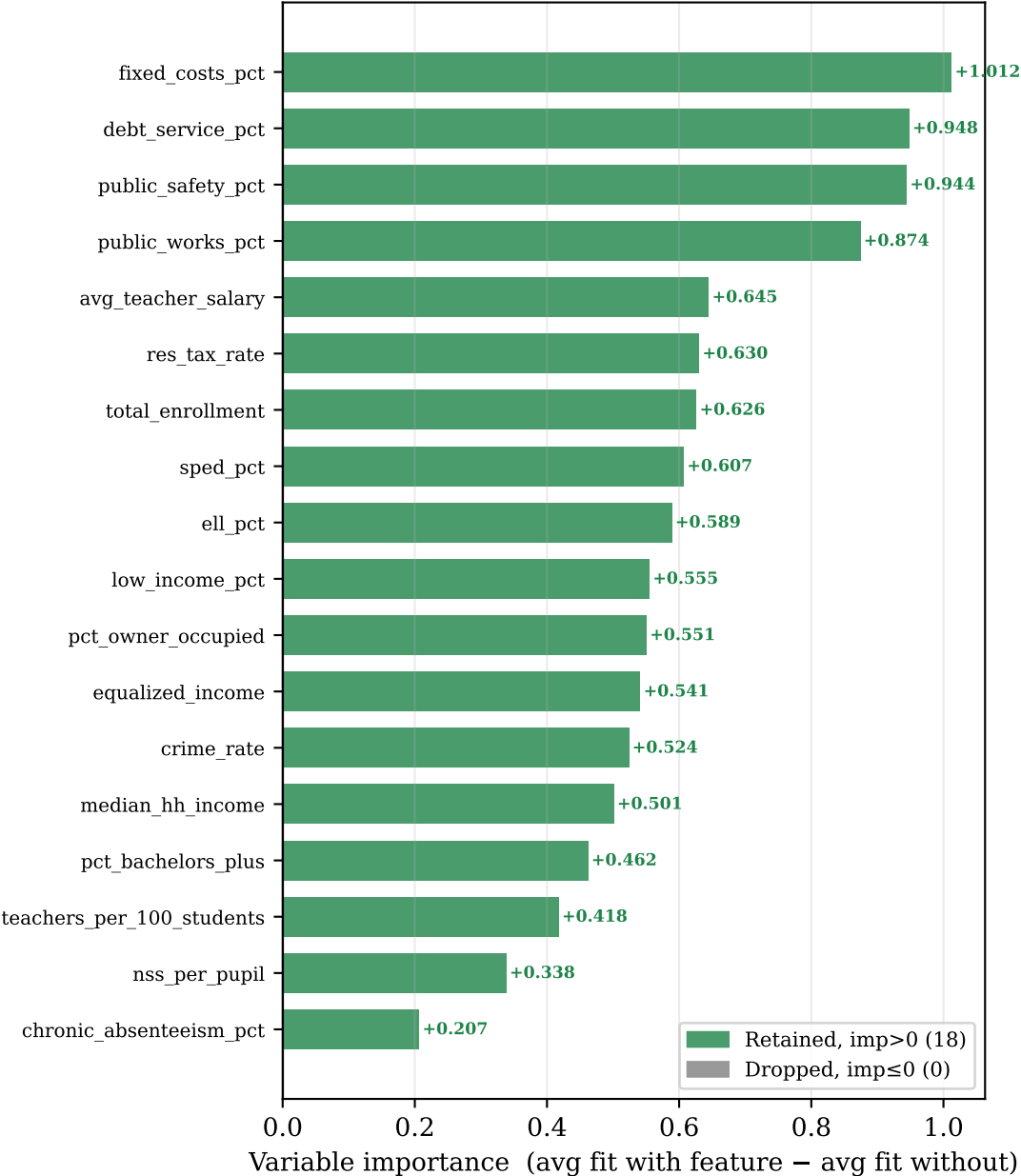
The Takeaway

Chronic absenteeism is again a top driver, and Saugus's rate (31.2%) remains far above the towns it's compared with here. Saugus is overcoming a real headwind on this measure — its grades 3-8 academic base, teacher compensation, and enrollment scale appear to be the offsetting strengths that let it outperform despite high absenteeism.

Variable Importance (Exhibit 5): Education Budget Share

18 candidates · n_random=1000 · 18 retained (imp > 0) · 0 with imp ≤ 0 (benign or harmful)

All Candidates — Variable Importance
Green = helps (>0) · Grey ≈ 0 benign · Red < 0 harmful

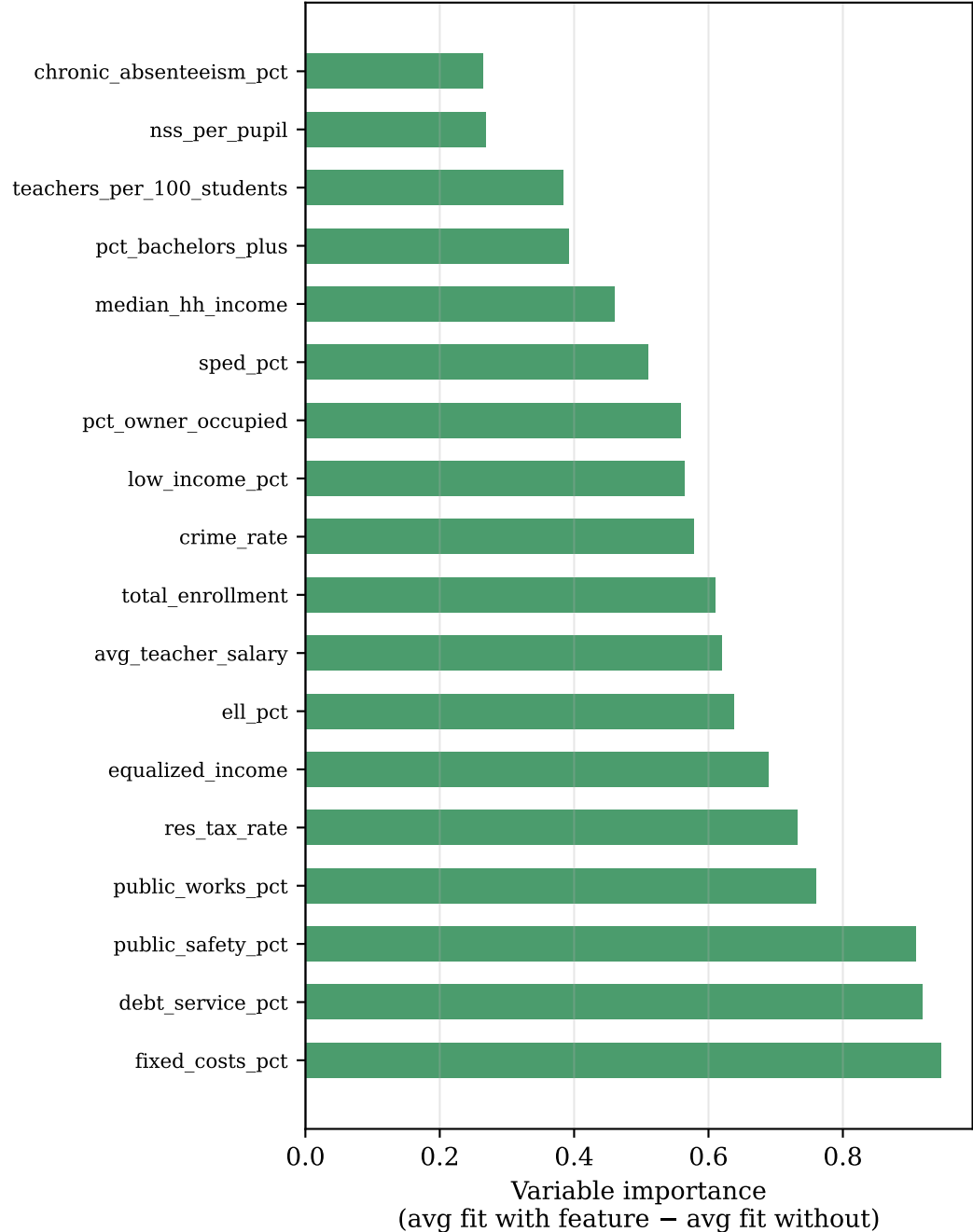


Variable importance: avg adjusted fit of grid cells containing feature – avg fit of cells without it. Univariate LOO r: single-feature leave-one-out correlation.

Saugus RBP Analysis: Education Budget Share

Prediction task = Saugus · Predicted: 36.5pp Actual: 32.3pp Gap: -4.2pp (Fit: 0.2001)

Variable Importance (Exhibit 5)



Most Relevant Towns (highest weight)

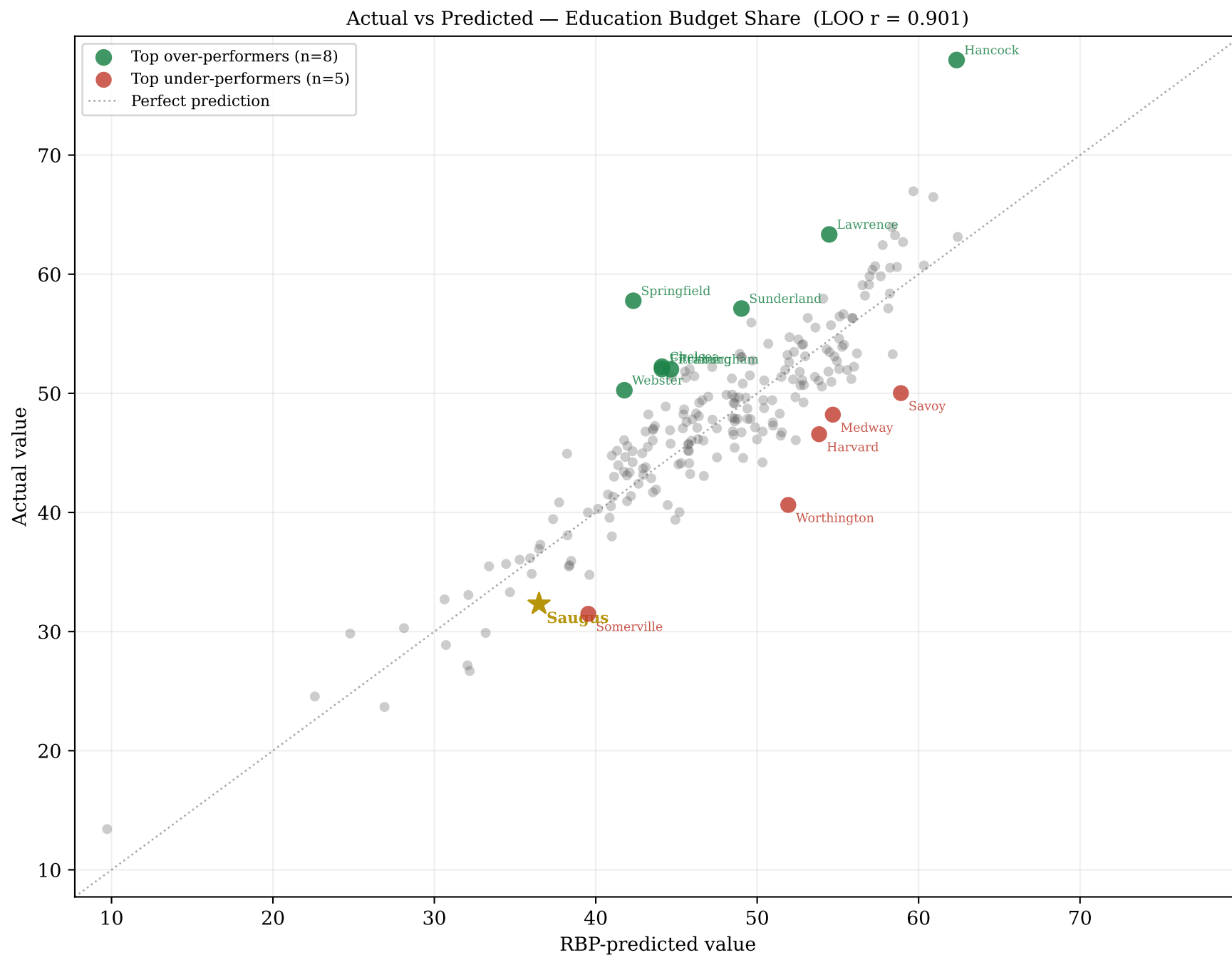
Town	Weight	ed_budget_sh
Gloucester	0.0497	33.30
Pittsfield	0.0482	40.53
Petersham	0.0466	56.65
Everett	0.0410	41.38
Gardner	0.0396	43.68

Least Relevant Towns (lowest weight)

Town	Weight	ed_budget_sh
Sunderland	-0.0205	57.12
Shutesbury	-0.0190	60.61
Amherst	-0.0179	52.21
Boxford	-0.0148	66.48
Topsfield	-0.0138	59.83

Over- and Under-Performers: Education Budget Share

Saugus gap: -4.2pp · Gold star = Saugus · Green = over-performers · Red = under-performers



Residual = actual – predicted. Positive = performing better than demographics suggest.

What Overachievers Do Differently: Education Budget Share

Feature values for top overachievers vs Saugus — ordered by RBP variable importance

Feature values — Saugus (highlighted) vs overachievers
Ordered by |importance|, top 15 shown

Feature	Saugus	Hancock	Springfie	Lawrence	Imp
fixed_costs_pct	25.3	2.43	19.7	7.45	+0.946
debt_service_pct	8.63	0.00	3.32	3.65	+0.918
public_safety_pct	15.2	4.61	11.0	8.66	+0.909
public_works_pct	7.35	6.46	1.35	3.98	+0.760
res_tax_rate	10.7	2.79	16.1	9.25	+0.732
equalized_income	1.17B	13M	2.89B	1.77B	+0.690
ell_pct	13.0	0.00	17.1	42.1	+0.639
avg_teacher_salary	89,778	59,463	84,692	84,516	+0.620
total_enrollment	3,084	106.0	29,468	15,500	+0.610
crime_rate	2,813	0.00	8,923	3,435	+0.579
low_income_pct	46.6	36.2	85.0	85.1	+0.565
pct_owner_occupied	73.9	77.8	49.5	30.3	+0.559
sped_pct	18.8	24.1	26.8	19.4	+0.510
median_hh_income	100,819	93,125	51,339	57,903	+0.461
pct_bachelors_plus	31.4	37.8	19.6	16.4	+0.392

Overachiever residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	32.3	36.5	-4.2pp
Hancock	78.0	62.4	+15.6pp
Springfield	57.8	42.3	+15.4pp
Lawrence	63.3	54.5	+8.9pp
Webster	50.3	41.8	+8.5pp
Chelsea	52.2	44.1	+8.1pp
Sunderland	57.1	49.0	+8.1pp

Gap = actual – predicted. Positive = outperforming demographic expectation. Feature values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Education Budget Share

Predicted 36.5pp · Actual 32.3pp · Gap -4.2pp (higher is better) · Saugus ranks 202 of 219 MA districts on this measure

The Bottom Line

Saugus directs 32.3% of its municipal budget to education, 3.9pp below its predicted 36.2% — one of the largest shortfalls in the state (199th of 219 districts, bottom 10%). Unlike the academic measures, this gap reflects a municipal budgeting choice, not a demographic constraint.

What Shapes Saugus's Predicted Peer Group

Top RBP Exhibit 5 drivers of the Saugus prediction, vs. the statewide median:

Factor	Saugus	MA median	Importance
Pension & benefits share of budget	25.3%	17.5%	+0.95
Debt service share of town budget	8.6%	5.0%	+0.92
Police & fire share of town budget	15.2%	11.8%	+0.91
Public works share of town budget	7.3%	5.3%	+0.76
Residential tax rate (per \$1,000)	10.7	12.6	+0.73

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Pension & benefits share of budget	25.3%	17.6%	-7.7pp
Debt service share of town budget	8.6%	3.0%	-5.6pp
Police & fire share of town budget	15.2%	10.7%	-4.5pp
Public works share of town budget	7.3%	4.0%	-3.3pp
Residential tax rate (per \$1,000)	10.7	12.3	+1.7

Difference = over-performer median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

All four top local drivers are competing budget categories where Saugus spends more than the state median: debt service (8.6% vs. 5.0%), pensions & benefits (25.3% vs. 17.5%), police & fire (15.2% vs. 11.8%), and public works (7.4% vs. 5.3%). Together these fixed costs crowd out the education share. Of the four outcomes, this is the one where the lever sits with Town Meeting and municipal budgeting, not the school district.

Combined Model Summary: All Four Outcomes

RBP factor selection results across MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, and Education Budget Share

Model	Predicted	Actual	Gap	Direction	LOO r	Features
MCAS Grades 3–8	22.3%	29.0%	+6.7pp	above	+0.945	17
Dropout Rate	3.6%	2.9%	-0.7pp	below	+0.839	17
MCAS Grade 10 (ELA)	42.3%	47.0%	+4.7pp	above	+0.899	17
Education Budget Share	36.5%	32.3%	-4.2pp	below	+0.901	18

Feature Cross-Reference — Which Factors Earn Their Place Across Models

A feature appearing in multiple models is a robust signal, not a model-specific artifact.

Feature	# Models	MCAS 3-8	Dropout	MCAS 10	Ed Budget
total enrollment	4	+0.310	+0.493	+0.316	+0.610
chronic absenteeism pct	4	+0.379	+0.886	+0.165	+0.265
equalized income	4	+0.332	+0.378	+0.231	+0.690
ell pct	4	+0.267	+0.471	+0.245	+0.639
crime rate	4	+0.289	+0.435	+0.213	+0.579
avg teacher salary	4	+0.282	+0.374	+0.212	+0.620
pct bachelors plus	4	+0.356	+0.488	+0.235	+0.392
res tax rate	4	+0.276	+0.235	+0.226	+0.732
median hh income	4	+0.233	+0.527	+0.231	+0.461
pct owner occupied	4	+0.333	+0.385	+0.159	+0.559
low income pct	4	+0.257	+0.432	+0.181	+0.565
sped pct	4	+0.188	+0.374	-0.006	+0.510
nss per pupil	4	+0.247	+0.352	+0.192	+0.269
teachers per 100 students	4	+0.146	+0.347	+0.133	+0.384
attending pct	3	+0.189	+0.164	+0.023	—
avg mcas	2	—	+0.523	+0.293	—
dropout pct	2	+0.228	—	+0.156	—
fixed costs pct	1	—	—	—	+0.946
debt service pct	1	—	—	—	+0.918
public safety pct	1	—	—	—	+0.909
public works pct	1	—	—	—	+0.760
mcas10 ela	1	—	+0.490	—	—
ed budget share	1	+0.069	—	—	—

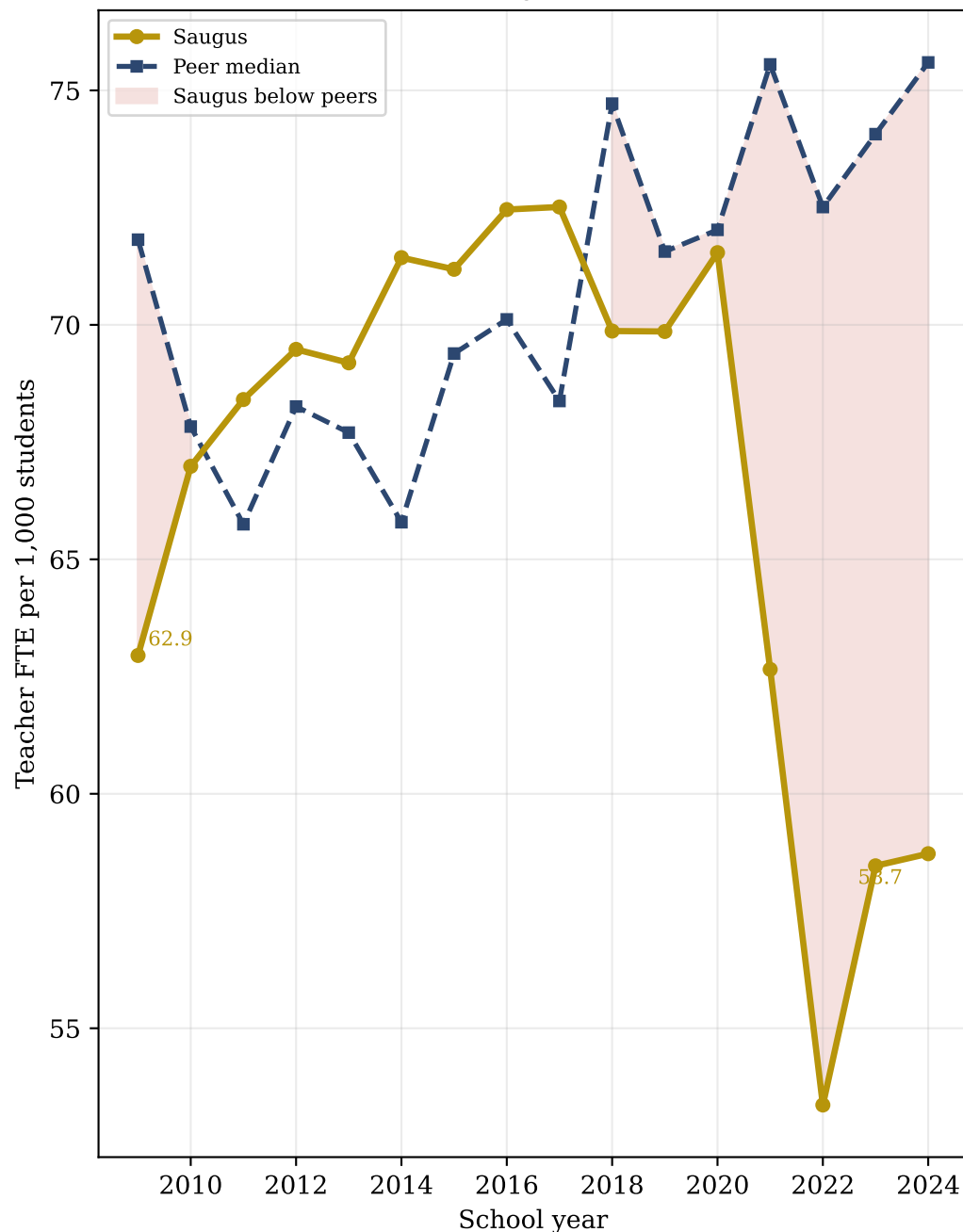
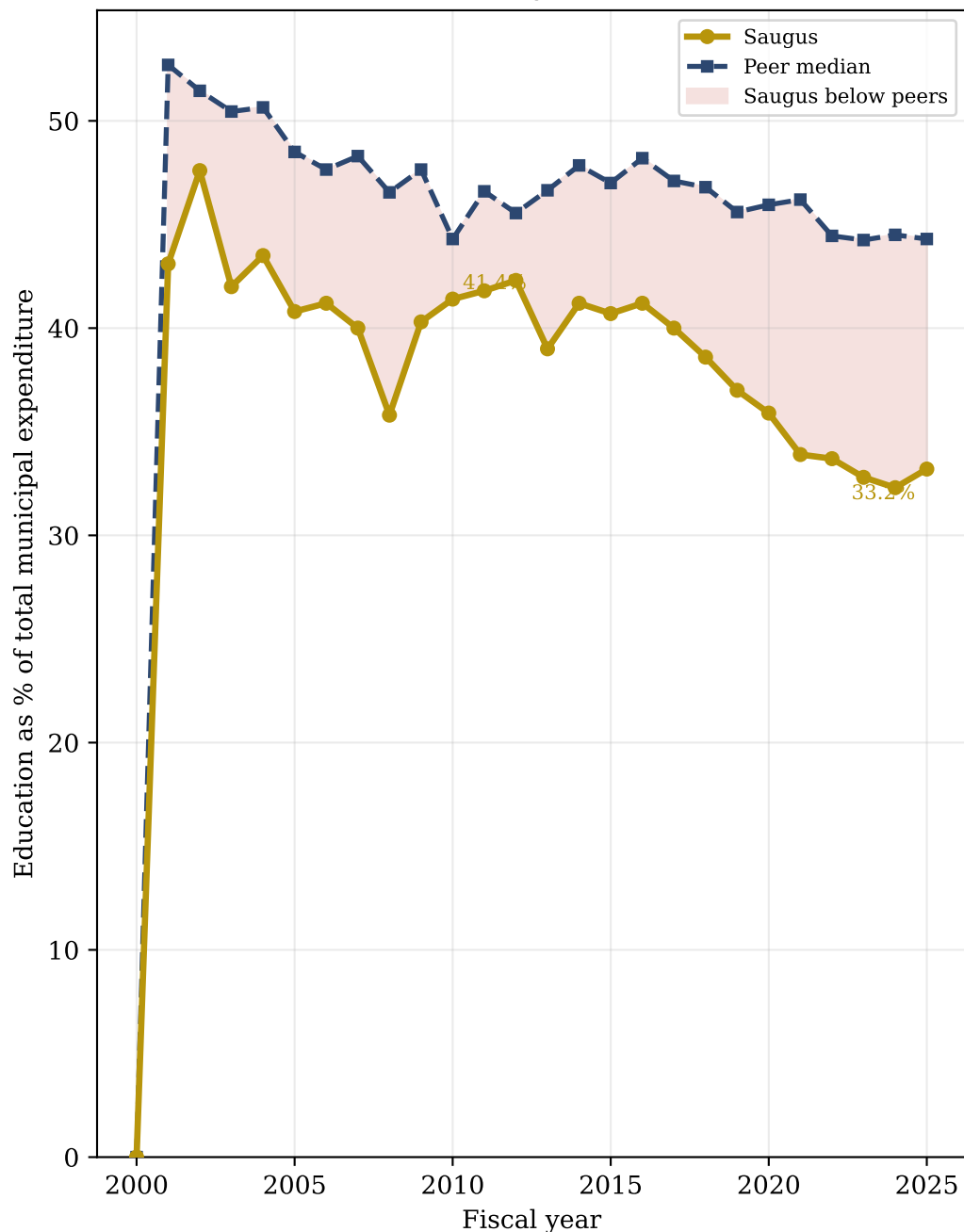
Green rows = feature appears in 3+ models. Cell colour: green = positive importance, red = negative importance, white = not in that model (—).

Budget Allocation & Teacher Density — 15-Year Trajectory

Source: MA DLS Schedule A (budget) · MA DESE Staffing (teachers) · Peer comparison: 8 demographically similar overachiever towns

Education Budget Share
(Schedule A, general fund)

Teacher Density
(DESE Staffing + Enrollment)



Independent cross-validation: Ridge regression on 221 MA districts ($R^2=0.83$) identifies chronic absenteeism as the #1 predictor of MCAS outcomes (importance=3.15), confirming the RBP Exhibit 5 finding by a separate method. Peer set: 8 demographically similar overachiever towns (Mahalanobis $d_M \leq 2.0$).

Optimum Profile — What Overachievers Look Like

Each grid shows actionable feature values for Saugus vs overachiever towns. Showing both peer groups tests whether the finding holds across definitions.

Overachievers — all 20 MA districts that beat their demographic prediction

Feature	Saugus	Winchendo	Lawrence	Springfie	Chelsea	North Ada	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	35.6%	29.2%	32.4%	35.4%	30.4%	23.8%	-7.4pp	20
Teachers / 100 students	5.9	6.7	7.4	7.4	6.7	8.7	7.5	+1.6	20
Avg teacher salary	\$89,778	\$83,722	\$84,516	\$84,692	\$89,573	\$76,435	\$87,136	\$-2,642	20
Net school spending / pupil	\$17,369	\$19,559	\$23,178	\$26,616	\$22,535	\$26,299	\$20,975	+\$3,606	20
Shown: Winchendon, Lawrence, Springfield, Chelsea, North Adams (+ 15 more in median)									

Demographically Similar Overachievers — 16 towns within 2σ Mahalanobis distance of Saugus on income, property wealth, poverty, size, education & ELL

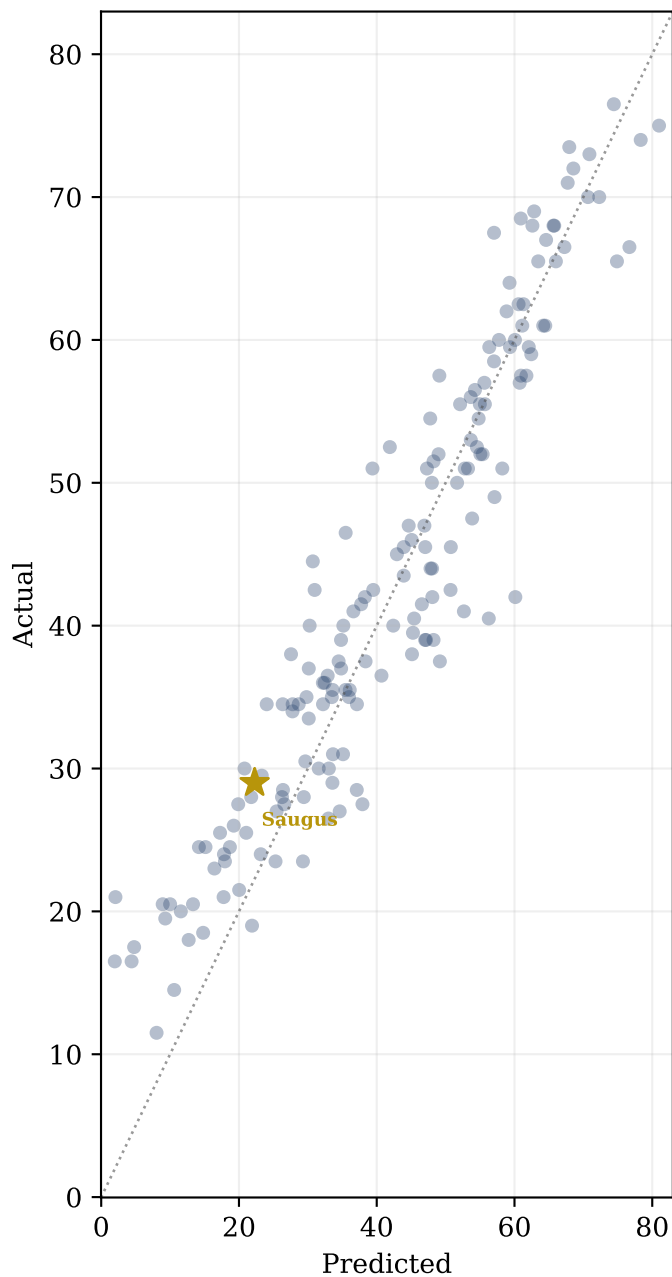
Feature	Saugus	Norwood	Clinton	Oxford	Westport	Fairhaven	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	25.1%	7.7%	19.9%	13.8%	15.6%	21.2%	-10.0pp	16
Teachers / 100 students	5.9	8.8	7.3	7.6	7.5	7.5	7.6	+1.7	16
Avg teacher salary	\$89,778	\$92,324	\$87,059	\$80,053	\$83,017	\$82,224	\$87,136	\$-2,642	16
Net school spending / pupil	\$17,369	\$21,084	\$18,512	\$18,176	\$18,099	\$17,287	\$20,213	+\$2,844	16
Shown: Norwood, Clinton, Oxford, Westport, Fairhaven (+ 11 more in median)									

Similarity metric: Mahalanobis distance $d_M = \sqrt{(\Delta x \cdot \Sigma^{-1} \cdot \Delta x')}$, Σ = sample covariance of all ~169 MA districts on (median_hh_income, equalized_income, low_income_pct, total_enrollment, pct_bachelors_plus, ell_pct). Threshold: $d_M \leq 2.0$ (2σ). Pre-specified; not tuned to result. Gap = peer median – Saugus. Red = Saugus below peer median, Green = at or above.

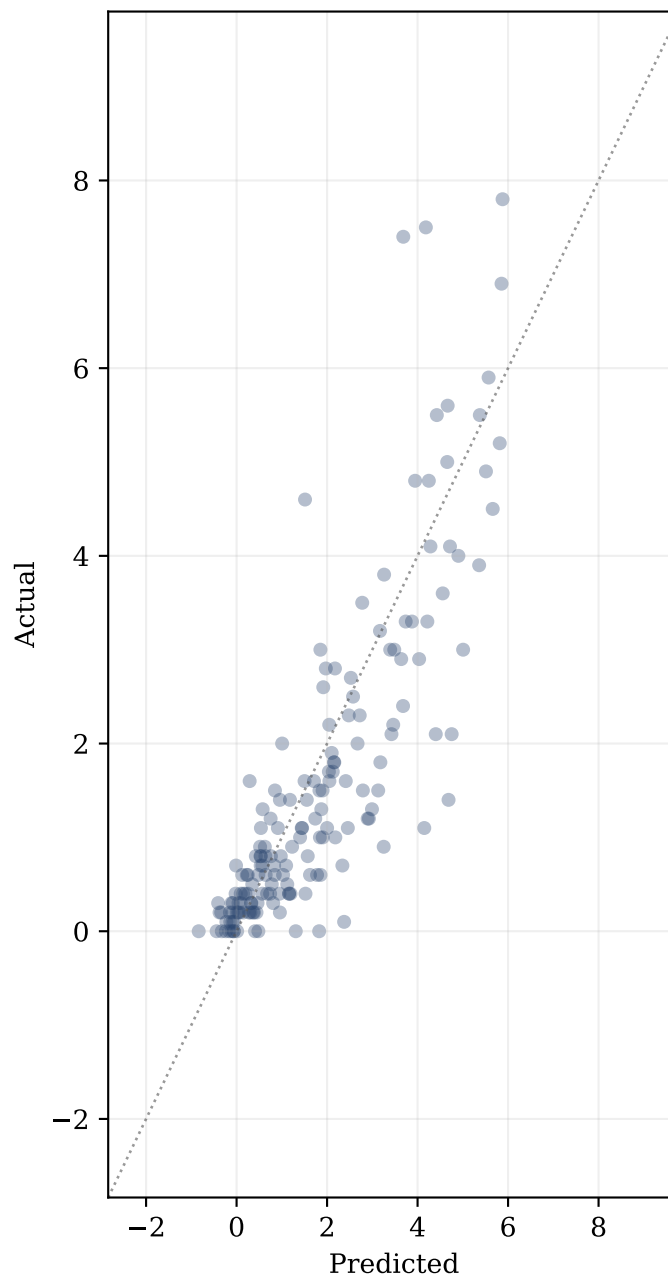
Actual vs. Predicted: All Three Models

Each dot = one MA district. Saugus highlighted in gold.

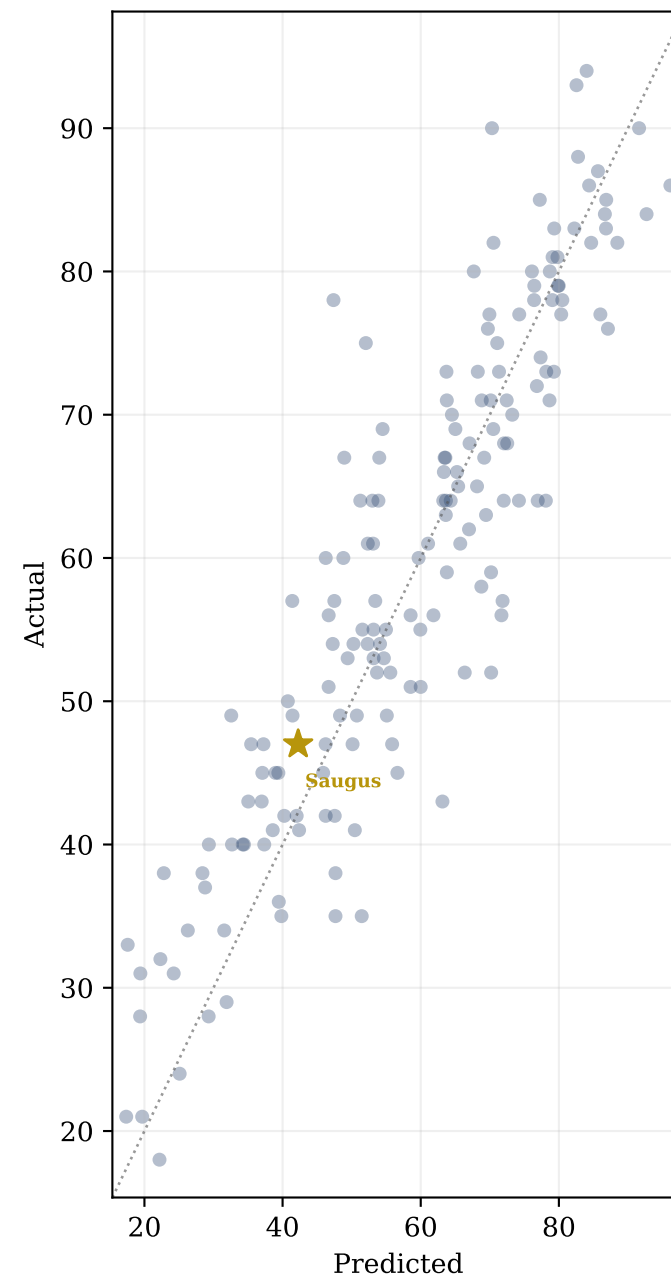
MCAS Grades 3-8
LOO r = 0.945



Dropout Rate
LOO r = 0.839



MCAS Grade 10 (ELA)
LOO r = 0.899



Leave-one-out predictions — each district excluded from its own prediction.